Supplementary Material

On Climate Variability and Civil War in Asia

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Published in *Climatic Change* (2014), doi 10.1007/s10584-013-1004-0

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I. Variable descriptions

I.I. Grid-level

The grid-level dataset was constructed from v.1.01 of PRIO-GRID. See Tollefsen, Strand, and Buhaug (1) and documentation found at http://www.prio.no/CSCW/Datasets/PRIO-Grid/ for further information.

I.I.I. Dependent variable

Civil war onset – dichotomous variable indicating whether a cell experiences a civil war onset in a given year (2,3). The point coordinates for the exact location of civil war onset are based on geo-coding. There are 87 valid onsets in the gridded analysis.

I.I.II. Climate variables

All climate variables include a current-year (t) and one-year lagged version (t-1) to account for the possibility of a delay in the effect on conflict risk. In order to ensure that the individual components in the interaction terms work in the same direction (i.e., higher values denoting expected higher conflict risk), the precipitation growth and deviation components were inversed when joined with the conditional variable.

Prec – absolute values give the yearly total amount of precipitation (in millimeter) in the cell, based on monthly meteorological statistics from the University of Delaware[4]. Values have been rescaled for interpretability (per 1,000 mm).

Temp – gives the yearly mean temperature (in degrees Celsius) in the cell, based on monthly meteorological statistics from the University of Delaware (4).

Prec growth – measures precipitation variation between consecutive years: $(\text{precipitation}_t\text{-precipitation}_{t-1})/\text{precipitation}_{t-1}$.

Temp growth – measures temperature variation between consecutive years: $(\text{temperature}_t\text{-temperature}_{t-1})/\text{temperature}_{t-1}$. Absolute values have been converted to Kelvin before variable creation. Values have been rescaled (multiplied by 100) for interpretability.

Prec dev – measures precipitation deviation from 1961-90 mean, in standard deviations. Variable is bounded at the 1st and 99th percentile to avoid extreme values.

Temp dev – measures temperature deviation from 1961-90 mean, in standard deviations. Variable is bounded at the 1st and 99th percentile to avoid extreme values. Absolute values have been converted to Kelvin before variable creation.

Drought – is a dichotomous variable, indicating a drought event within a given year. It is based on an aggregated yearly Standardized Precipitation Index (SPI6) that captures within-year deviations in precipitation based on monthly data (see Codebook for PRIO-GRID for a more detailed description).

I.I.III. Control variables

Border distance (ln) – distance from the cell centroid to the nearest international border (or the coast for island states) in kilometer. Values have been rescaled (natural logarithm) to account for an expected non-linear effect.

Capital distance (ln) – distance from the cell centroid to the national capital in kilometer. Values have been rescaled (natural logarithm) to account for an expected non-linear effect.

Irrigated cropland (>10%) – dichotomous variable, coded 1 if a cell's land area includes more than 10% irrigated cropland. Information on land cover is based on the Globcover 2009 project (5).

Rainfed cropland (>10%) - dichotomous variable, coded 1 if a cell's land area includes more than 10% rainfed cropland. Information on land cover is based on the Globcover 2009 project (5).

Cropland – dichotomous variable used in interaction models. Indicates whether more than 5% of a cell's area is used as cropland.

Local political exclusion (t-1) – dichotomous variable indicating whether a cell is inhabited by at least one ethnic group that is excluded from national politics. Data on the spatial distribution of ethnic groups is taken from the GeoEPR (6) dataset and has been combined with information on groups' political status from the EPR-ETH (7) dataset. Politically irrelevant, small (<2.5% of the total population), and geographically dispersed groups are excluded from the dataset. Groups coded 'discriminated', 'irrelevant', 'powerless', 'regional autonomy' or 'separatist' are considered politically marginalized.

Local population (ln) – measure of total population living in a cell in a given year. Data on the years 1990, 1995, 2000 and 2005 is derived from the Gridded Population of the World, version 3 (8). The remaining years are interpolated and extrapolated assuming linear growth in the natural logarithm of the population size. Values have been rescaled (natural logarithm) to account for an expected non-linear effect.

***Local conflict* history** – cell-specific decay function on conflict history, based on conflict site data. This control variable is given by 2 to the power of –(number of years since last civil war or independence/α), where α represents the half-life parameter. A half-life of two years gave the best fit to the conflict data.

Sample specifications

The grid-level models are run on a reduced sample that includes all onset-cells plus a five percent random draw of all non-conflict observations. The 'most likely' subsample used in Models A145-A150 includes all cells with at least 5 % of agricultural land and are inhabited by at least one politically marginalized ethnic group.

I.II. Country-level

The dependent variable, the climate variables as well as the indicators on land-usage are aggregated from the gridded dataset to a country-level format.

I.II.I. Dependent variable

Civil war onset – dichotomous variable indicating whether a country experiences a civil war onset in a given year (2,3). There are 80 onset-years in the country-level dataset.

I.II.II. Climate variables

All climate variables include a lagged version to account for the possibility of a time-variant effect on conflict risk. In order to ensure that the individual components in the interaction terms work in the same direction (i.e., higher values denoting expected higher conflict risk), the precipitation growth and deviation components were inversed when joined with the conditional variable.

Prec – absolute values give the area-weighted yearly total amount of precipitation (in millimeter) in the country, based on monthly gridded meteorological statistics from the University of Delaware (4). Values have been rescaled for interpretability.

Temp – gives the area-weighted yearly mean temperature (in degrees Celsius) in the country, based on monthly gridded meteorological statistics from the University of Delaware (4).

Prec growth **–** measures precipitation variation between consecutive years: $(\text{precipitation}_t\text{-precipitation}_{t-1})/\text{precipitation}_{t-1}$.

Temp growth **–** measures temperature variation between consecutive years: $(\text{temperature}_t\text{-temperature}_{t-1})/\text{temperature}_{t-1}$. Absolute values have been converted to Kelvin before variable creation. Values have been rescaled (multiplied by 100) for interpretability.

Prec dev **–** measures precipitation deviation from the 1961-90 mean, in standard deviations.

Temp dev **–** measures temperature deviation from 1961-90 mean, in standard deviations. Absolute values have been converted to Kelvin before variable creation.

Drought **–** area-weighted, country-aggregated variable indicating the percentage share of country-area affected by a drought in a given year.

I.II.III. Control Variables

Rainfed cropland – area-weighted, country-aggregated variable indicating the percentage share of rainfed cropland in a country.

Irrigated cropland – area-weighted, country-aggregated variable indicating the percentage share of irrigated cropland in a country.

Cropland – dichotomous variable used in interaction models. Indicates whether more than 20% of a country's area is used as cropland.

Population (ln) – country population. Data derived from the Penn World Tables 7.0 (11). Missing data is replaced with data from the World Development Indicators (13). Values have been rescaled (natural logarithm) to account for an expected non-linear effect.

Political exclusion (t-1) – country-level indicator of extent of ethnopolitical exclusion in a country. The variable is bounded within the interval [0, 1] and gives the ratio of the population for the largest excluded ethnic group relative to the ethnic group(s) in power, based on the EPR-ETH dataset (7).

Democracy (t-1) – country-level measure of political regime type, based on the Xpolity index (9). The index excludes the Polity (10) components on political participation and goes from -6 (autocracy) to +7 (democracy).

GDP/capita (ln, t-1) – country-level indicator of GDP/capita. Data is derived from the Penn World Tables 7.0 (11). Missing data are replaced with Expanded Trade and GDP Data (version 4.1) (12) in the reference year (2000), based on the mean ratio between the two indicators. Values are interpolated and extrapolated for the remaining years, and rescaled (natural logarithm) to account for an expected non-linear effect.

Food production – country-level indicator of food production per capita, based on value of gross production in constant 2004–06 I$. Data derived from FAOSTAT (14). *Food growth* measures the percentage change in food production per capita between *t*–1 and *t*.

Year – the calendar year of observation, included in some models to capture possible common, linear time trends.

Local conflict history – country-specific decay function on conflict history. This control variable is given by 2 to the power of –(number of years since last civil war or independence/α), where α represents the half-life parameter. A half-life of two years gave the best fit to the conflict data.

I.III. Data sources

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II. Descriptive Statistics

Table 1 Descriptive statistics, grid-level, 1950-2008

Variable	# of Obs		Mean		Std. Dev		Min		Max	
	100%	5%	100%	5%	100%	5%	100%	5%	100%	5%
Prec	554,131	24,917	.895	.835	.813	.784	.067	0.67	15.005	13.552
Prec (t-1)	554,131	24,917	.896	.837	.814	.788	.067	0.69	15.005	14.532
Temp	554,131	24,917	13.089	11.960	11.087	10.942	-17.508	-16.133	31.017	30.575
Temp (t-1)	554,131	24,917	13.077	11.960	11.096	10.949	-17.508	-16.767	31.017	30.575
Prec growth (t-1 to t)	554,131	24,917	.032	.030	.279	.281	-.904	-.830	6.069	5.392
Prec growth (t-1 to t, inv)	554,131	24,917	-.032	-.030	.279	.281	-6.069	-5.392	.904	.830
Prec growth (t-2 to t-1)	554,131	24,917	.033	.035	.281	.284	-.903	-.871	6.069	5.482
Prec growth (t-2 to t-1, inv)	554,131	24,917	-.033	-.035	.281	.284	-6.069	-5.482	.903	.871
Temp growth (t-1 to t)	554,131	24,917	.005	.002	.240	.245	-5.366	-1.664	5.256	2.328
Temp growth (t-2 to t-1)	554,131	24,917	.006	.010	.238	.246	-5.366	-1.795	5.256	3.724
Prec dev	554,131	24,917	.018	.022	1.055	1.064	-2.371	-2.371	2.994	2.994
Prec dev (inv)	554,131	24,917	-.018	-.022	1.055	1.064	-2.994	-2.994	2.371	2.371
Prec dev (t-1)	554,131	24,917	.027	.036	1.055	1.072	-2.371	-2.371	2.994	2.994
Prec dev (t-1, inv)	554,131	24,917	-.027	-.036	1.055	1.072	-2.994	-2.994	2.371	2.371
Temp dev	554,131	24,917	.310	.299	1.192	1.200	-2.539	-2.539	3.485	3.485
Temp dev (t-1)	554,131	24,917	.292	.294	1.183	1.196	-2.539	-2.539	3.485	3.485
Drought	541,646	24,303	.185	.182	.388	.386	0	0	1	1
Drought (t-1)	533,214	23,876	.187	.186	.390	.389	0	0	1	1
Civil war onset	569,441	25,580	.00015	.003	.012	.058	0	0	1	1
Distance to border (ln)	569,441	25,580	.754	.757	.187	.185	0	0	1	1
Distance to capital (ln)	569,441	25,580	.867	.868	.092	.092	.274	.277	1	1
Rainfed Cropland	569,441	25,580	.341	.331	.474	.471	0	0	1	1
Irrigated Cropland	569,441	25,580	.171	.153	.377	.360	0	0	1	1
Cropland	569,441	25,580	.708	.699	.455	.459	0	0	1	1
Local population (ln)	569,273	25,572	10.493	10.372	2.538	2.559	0	0	18.942	16.696
Local political exclusion (t-1)	567,981	25,507	.472	.477	.499	.499	0	0	1	1
Democracy (t-1)	569,441	25,580	1.778	1.664	3.118	2.955	-6	-6	7	7
GDP/capita (ln, t-1)	569,343	25,572	6.187	6.191	1.590	1.644	3.375	3.375	10.985	10.976
Year	569,441	25,580	1980.427	1979.981	17.166	17.238	1950	1950	2008	2008
Local conflict history	569,441	25,580	.191	.107	.344	.239	1.86e-09	1.86e-09	1	1

Table 2 Descriptive statistics, country-level, 1950-2008

Variable	Obs.	Mean	Std. Dev.	Min	Max
Prec	1,387	1.368	.732	.145	3.809
Prec (t-1)	1,367	1.366	.730	.155	3.809
Temp	1,387	17.774	8.388	-1.609	27.834
Temp (t-1)	1,367	17.761	8.395	-1.609	27.834
Prec growth (t-1 to t)	1,367	.011	.148	-.587	.684
Prec growth (t-1 to t, inv)	1,367	-.011	.148	-.684	.587
Prec growth (t-2 to t-1)	1,347	.011	.148	-.587	-.587
Prec growth (t-2 to t-1, inv)	1,347	-.011	.148	-.684	.587
Temp growth (t-1 to t)	1,367	.005	.170	-.861	.859
Temp growth (t-2 to t-1)	1,347	.008	.171	-.861	.859
Prec dev	1,380	.017	1.076	-4.413	5.418
Prec dev (inv)	1,380	-.017	1.076	-5.418	4.413
Prec dev (t-1)	1,361	.017	1.069	-4.413	5.418
Prec dev (t-1, inv)	1,361	-.017	1.069	-5.418	4.413
Temp dev	1,380	.435	1.269	-2.738	5.669
Temp dev (t-1)	1,361	.417	1.265	-3.528	5.669
Drought	1,355	.186	.224	0	1
Drought (t-1)	1,325	.188	.225	0	1
Civil war onset	1,387	.0577	.233	0	1
Rainfed cropland	1,387	7.748	11.818	0	55.847
Irrigated cropland	1,387	10.842	8.832	.101	39.206
Cropland	1,387	.660	.474	0	1
Population (ln)	1,387	9.966	1.871	5.099	14.091
Political exclusion (t-1)	1,315	.141	.192	0	.861
Democracy (t-1)	1,387	1.409	3.636	-6	7
GDP/capita (ln, t-1)	1,380	6.618	1.604	3.375	10.985
GDP/capita (ln, t-1, inv)	1,380	.160	.039	.091	.296
Year	1,387	1981.417	16.952	1950	2008
Conflict history	1,387	.381	.436	1.86e-09	1

III. Models documented in article

Table 3 Country-level regression models in Table 1

	Logit			*FE Logit*	*FE OLS*	
	(1*a*)	(2*a*)	(3*a*)	(4*a*)	(5*a*)	(6*a*)
VARIABLES	Civil war onset	Civil war onset	Civil war onset	Civil war onset	Civil war onset	Civil war onset
Prec growth (t-2 to t-1)	1.345					
	(0.783)					
Temp growth (t-2 to t-1)	-0.369					
	(0.811)					
Prec dev (t-1)		-0.091				
		(0.087)				
Temp dev (t-1)		-0.002				
		(0.126)				
Drought (t-1)			1.428*			
			(0.563)			
Prec (t-1)				-0.334	-0.016	-0.007
				(0.697)	(0.032)	(0.032)
Temp (t-1)				-0.222	-0.010	0.011
				(0.333)	(0.013)	(0.015)
Irrigated cropland	-0.006	-0.007	-0.005			
	(0.012)	(0.012)	(0.013)			
Rainfed cropland	0.008	0.008	0.009			
	(0.014)	(0.015)	(0.015)			
Population (ln)	0.239	0.239	0.242			
	(0.126)	(0.123)	(0.124)			
Political exclusion (t-1)	-1.162	-1.163	-0.985			
	(0.939)	(0.957)	(0.987)			
Democracy (t-1)	0.139**	0.142**	0.145**			
	(0.040)	(0.040)	(0.038)			
GDP/capita (ln, t-1)	-0.750**	-0.748*	-0.773**			
	(0.281)	(0.293)	(0.284)			
Year	0.053*	0.052	0.057*	0.020*	0.001*	
	(0.023)	(0.027)	(0.024)	(0.009)	(<0.001)	
Conflict history	0.827*	0.805*	0.764	-0.877*	-0.035	-0.135**
	(0.401)	(0.406)	(0.415)	(0.411)	(0.020)	(0.024)
Constant	-106.502*	-104.549*	-114.491*		-1.438	-0.082
	(43.980)	(52.456)	(45.823)		(0.786)	(0.267)
Observations	1,251	1,251	1,251	837	1,367	1,367
R-squared					0.006	0.073
Log Likelihood	-247.3	-248.4	-245.9	-200.5		

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Models 4*a*–6*a* include country fixed effects; Model 6*a* include country-specific time trends

Table 4 Grid-level regression models in Table 1

	Logit			*FE Logit*	*FE OLS*	
	(1*b*)	(2*b*)	(3*b*)	(4*b*)	(5*b*)	(6*b*)
VARIABLES	Civil war onset	Civil war onset	Civil war onset	Civil war onset	Civil war onset	Civil war onset
Prec growth (t-2 to t-1)	0.515					
	(0.382)					
Temp growth (t-2 to t-1)	-0.452					
	(0.554)					
Prec dev (t-1)		0.061				
		(0.090)				
Temp dev (t-1)		-0.051				
		(0.156)				
Drought (t-1)			0.338			
			(0.239)			
Prec (t-1)				0.225	0.001*	0.001*
				(0.149)	(0.001)	(0.001)
Temp (t-1)				-0.032	-0.000	>-0.001
				(0.021)	(<0.001)	(<0.001)
Border distance (ln)	-1.449*	-1.449	-1.435			
	(0.730)	(0.743)	(0.748)			
Capital distance (ln)	-0.072	-0.084	0.101			
	(2.089)	(2.108)	(2.083)			
Irrigated cropland (>10%)	-0.779*	-0.797*	-0.795*			
	(0.368)	(0.368)	(0.379)			
Rainfed cropland (>10%)	-0.708**	-0.706*	-0.719*			
	(0.273)	(0.274)	(0.281)			
Local population (ln)	0.467**	0.465**	0.464**			
	(0.091)	(0.094)	(0.094)			
Local political exclusion (t-1)	1.363*	1.363*	1.367*			
	(0.597)	(0.598)	(0.595)			
Democracy (t-1)	0.180*	0.185*	0.189*			
	(0.078)	(0.075)	(0.076)			
GDP/capita (ln, t-1)	-0.461	-0.466	-0.458			
	(0.285)	(0.272)	(0.279)			
Year	0.041	0.043	0.041*	0.016*	<0.001**	
	(0.021)	(0.022)	(0.021)	(0.007)	(<0.001)	
Local conflict history	3.313**	3.275**	3.244**	0.393	0.002*	>-0.001
	(0.559)	(0.567)	(0.558)	(0.282)	(0.001)	(0.001)
Constant	-89.838*	-93.187*	-90.767*		-0.098**	0.002
	(40.009)	(42.589)	(39.955)		(0.037)	(0.001)
Observations	23,356	23,356	23,356	25,547	30,845	30,845
R-squared					<0.001	0.005
Log Likelihood	-399.0	-400.2	-399.8	-491.9		

Robust standard errors in parentheses; ** p<0.01, * p<0.05
Models 4*b*–6*b* include country fixed effects; Model 6*b* include country-specific time trends

IV. Country-level analysis

Table 5 Rare events logistic regression

VARIABLES	(A1) Civil war onset	(A2) Civil war onset	(A3) Civil war onset	(A4) Civil war onset	(A5) Civil war onset	(A6) Civil war onset
Prec		-0.147				
		(0.304)				
Temp		0.095				
		(0.050)				
Prec (t-1)			-0.230			
			(0.309)			
Temp (t-1)			0.103*			
			(0.050)			
Prec growth (t-1 to t)				0.350		
				(0.644)		
Prec growth (t-2 to t-1)				1.381		
				(0.748)		
Temp growth (t-1 to t)				-1.223		
				(0.807)		
Temp growth (t-2 to t-1)				-0.909		
				(0.793)		
Prec dev					-0.004	
					(0.113)	
Prec dev (t-1)					-0.097	
					(0.089)	
Temp dev					-0.264*	
					(0.125)	
Temp dev (t-1)					0.084	
					(0.123)	
Drought						0.608
						(0.582)
Drought (t-1)						1.384*
						(0.569)
Irrigated cropland	-0.005	-0.017	-0.019	-0.005	-0.007	-0.003
	(0.012)	(0.014)	(0.014)	(0.012)	(0.012)	(0.012)
Rainfed cropland	0.009	-0.011	-0.011	0.008	0.009	0.011
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Population (ln)	0.231	0.350**	0.355**	0.232	0.236	0.237
	(0.124)	(0.125)	(0.129)	(0.124)	(0.123)	(0.123)
Political exclusion (t-1)	-0.945	-0.559	-0.546	-0.921	-0.893	-0.713
	(0.933)	(0.931)	(0.974)	(0.923)	(0.971)	(0.986)
Democracy (t-1)	0.137**	0.110**	0.109**	0.134**	0.138**	0.142**
	(0.039)	(0.039)	(0.039)	(0.039)	(0.040)	(0.036)
GDP/capita (ln, t-1)	-0.721**	-0.815**	-0.819*	-0.720**	-0.747*	-0.752**
	(0.277)	(0.315)	(0.326)	(0.278)	(0.315)	(0.278)
Year	0.051*	0.057*	0.057*	0.051*	0.058	0.057*
	(0.023)	(0.025)	(0.025)	(0.023)	(0.032)	(0.023)
Conflict history	0.784	0.309	0.294	0.781*	0.760	0.713
	(0.402)	(0.510)	(0.509)	(0.397)	(0.405)	(0.417)
Constant	-102.841*	-116.608*	-116.501*	-102.948*	-116.169	-113.776*
	(43.046)	(47.294)	(48.596)	(43.661)	(60.877)	(44.764)
Observations	1,251	1,251	1,251	1,251	1,251	1,251
Log Likelihood						

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 6 Logistic regression with interaction effect between GDP/capita and Prec growth

VARIABLES	(A7) Civil war onset	(A8) Civil war onset	(A9) Civil war onset	(A10) Civil war onset	(A11) Civil war onset
Prec growth (t-1 to t, inv)		0.100	4.060		
		(0.846)	(3.612)		
GDP x Prec growth (t-1 to t, inv)			-24.726		
			(22.873)		
Prec growth (t-2 to t-1, inv)				-1.428	-6.191
				(0.785)	(4.218)
GDP x Prec growth (t-2 to t-1, inv)					30.317
					(26.706)
GDP/capita (ln, t-1, inv)	20.827*	20.837*	20.770*	20.902*	21.385**
	(8.206)	(8.193)	(8.157)	(8.250)	(8.278)
Irrigated cropland	0.001	0.001	0.001	0.001	0.001
	(0.012)	(0.011)	(0.011)	(0.012)	(0.012)
Rainfed cropland	0.003	0.003	0.003	0.003	0.003
	(0.014)	(0.014)	(0.014)	(0.013)	(0.013)
Population (ln)	0.220	0.220	0.220	0.224	0.222
	(0.122)	(0.121)	(0.120)	(0.123)	(0.123)
Political exclusion (t-1)	-1.101	-1.104	-1.105	-1.097	-1.073
	(0.841)	(0.842)	(0.842)	(0.840)	(0.835)
Democracy (t-1)	0.112**	0.112**	0.113**	0.110**	0.110**
	(0.035)	(0.035)	(0.035)	(0.036)	(0.036)
Year	0.043*	0.043*	0.043*	0.043*	0.042
	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)
Conflict history	0.975*	0.975*	0.982*	0.994*	1.006*
	(0.406)	(0.406)	(0.411)	(0.401)	(0.408)
Constant	-93.669*	-93.751*	-94.614*	-93.624*	-92.841*
	(44.048)	(43.975)	(44.348)	(44.342)	(44.653)
Observations	1,251	1,251	1,251	1,251	1,251
Log Likelihood	-251.6	-251.5	-251.1	-250.1	-249.4

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 7 Logistic regression with interaction effect between GDP/capita and Temp growth

VARIABLES	(A12) Civil war onset	(A13) Civil war onset	(A14) Civil war onset	(A15) Civil war onset
Temp growth (t-1 to t)	-0.870	2.718		
	(0.885)	(3.104)		
GDP x Temp growth (t-1 to t)		-22.455		
		(16.196)		
Temp growth (t-2 to t-1)			-0.563	-7.631*
			(0.797)	(3.573)
GDP x Temp growth (t-2 to t-1)				44.784*
				(22.760)
GDP/capita (ln, t-1, inv)	20.823*	20.420*	20.896*	21.101*
	(8.205)	(8.269)	(8.253)	(8.440)
Irrigated cropland	0.001	0.001	0.001	0.001
	(0.012)	(0.012)	(0.012)	(0.012)
Rainfed cropland	0.003	0.003	0.003	0.003
	(0.014)	(0.014)	(0.014)	(0.014)
Population (ln)	0.221	0.221	0.219	0.224
	(0.122)	(0.122)	(0.122)	(0.121)
Political exclusion (t-1)	-1.082	-1.084	-1.107	-1.101
	(0.840)	(0.845)	(0.842)	(0.835)
Democracy (t-1)	0.112**	0.112**	0.112**	0.111**
	(0.036)	(0.036)	(0.036)	(0.036)
Year	0.043*	0.043*	0.043*	0.043*
	(0.022)	(0.022)	(0.022)	(0.022)
Conflict history	0.978*	0.992*	0.975*	1.007*
	(0.408)	(0.413)	(0.405)	(0.425)
Constant	-93.640*	-93.678*	-94.323*	-95.038*
	(44.073)	(44.223)	(44.449)	(44.889)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-251.0	-250.5	-251.3	-249.4

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 8 Logistic regression with interaction effect between GDP/capita and Prec dev

VARIABLES	(A16) Civil war onset	(A17) Civil war onset	(A18) Civil war onset	(A19) Civil war onset
Prec dev (inv)	0.001	0.166		
	(0.114)	(0.273)		
GDP x Prec dev (inv)		-1.029		
		(1.920)		
Prec dev (t-1, inv)			0.094	-0.608
			(0.087)	(0.459)
GDP x Prec dev (t-1, inv)				4.414
				(2.860)
GDP/capita (ln, t-1, inv)	20.830*	20.792*	20.882*	20.643*
	(8.160)	(8.148)	(8.207)	(8.335)
Irrigated cropland	0.001	0.001	<0.001	>-0.001
	(0.012)	(0.012)	(0.011)	(0.011)
Rainfed cropland	0.003	0.003	0.003	0.003
	(0.014)	(0.014)	(0.014)	(0.014)
Population (ln)	0.220	0.218	0.223	0.230
	(0.123)	(0.124)	(0.119)	(0.121)
Political exclusion (t-1)	-1.102	-1.155	-1.108	-0.899
	(0.842)	(0.851)	(0.858)	(0.881)
Democracy (t-1)	0.112**	0.112**	0.113**	0.114**
	(0.035)	(0.035)	(0.035)	(0.035)
Year	0.043*	0.043*	0.042	0.041
	(0.022)	(0.022)	(0.022)	(0.022)
Conflict history	0.975*	0.976*	0.967*	0.984*
	(0.412)	(0.413)	(0.409)	(0.421)
Constant	-93.656*	-94.271*	-91.765*	-89.850*
	(44.406)	(44.534)	(44.613)	(44.666)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-251.6	-251.5	-251.2	-250.3

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 9 Logistic regression with interaction effect between GDP/capita and Temp dev

VARIABLES	(A20) Civil war onset	(A21) Civil war onset	(A22) Civil war onset	(A23) Civil war onset
Temp dev	-0.241	-0.873*		
	(0.140)	(0.365)		
GDP x Temp dev		3.925		
		(2.197)		
Temp dev (t-1)			-0.002	-1.067**
			(0.127)	(0.375)
GDP x Temp dev (t-1)				6.654**
				(1.734)
GDP/capita (ln, t-1, inv)	22.518*	23.733*	20.844*	21.632*
	(9.423)	(9.402)	(8.856)	(9.054)
Irrigated cropland	>-0.001	-0.001	0.001	-0.001
	(0.012)	(0.012)	(0.012)	(0.011)
Rainfed cropland	0.003	0.003	0.003	0.004
	(0.013)	(0.013)	(0.014)	(0.014)
Population (ln)	0.219	0.225	0.220	0.230
	(0.124)	(0.124)	(0.122)	(0.119)
Political exclusion (t-1)	-1.056	-1.219	-1.100	-1.348
	(0.877)	(0.881)	(0.839)	(0.853)
Democracy (t-1)	0.116**	0.118**	0.112**	0.115**
	(0.036)	(0.036)	(0.036)	(0.036)
Year	0.054	0.058*	0.043	0.050
	(0.028)	(0.028)	(0.027)	(0.027)
Conflict history	0.954*	0.896*	0.975*	0.885*
	(0.405)	(0.411)	(0.404)	(0.400)
Constant	-116.559*	-124.348*	-93.894	-108.022*
	(57.758)	(56.751)	(53.922)	(53.952)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-249.5	-248.7	-251.6	-249.3

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 10 Logistic regression with interaction effect between GDP/capita and Drought

VARIABLES	(A24) Civil war onset	(A25) Civil war onset	(A26) Civil war onset	(A27) Civil war onset
Drought	0.705	3.055		
	(0.590)	(2.418)		
GDP x Drought		-15.194		
		(16.570)		
Drought (t-1)			1.406*	-1.005
			(0.558)	(1.875)
GDP x Drought (t-1)				15.479
				(11.047)
GDP/capita (ln, t-1, inv)	21.361**	25.835*	22.116**	17.311*
	(7.992)	(11.015)	(8.436)	(7.912)
Irrigated cropland	0.002	0.001	0.003	0.003
	(0.012)	(0.012)	(0.012)	(0.012)
Rainfed cropland	0.003	0.004	0.004	0.004
	(0.013)	(0.014)	(0.014)	(0.014)
Population (ln)	0.222	0.218	0.225	0.232*
	(0.122)	(0.123)	(0.120)	(0.118)
Political exclusion (t-1)	-1.025	-1.033	-0.926	-0.903
	(0.855)	(0.870)	(0.880)	(0.862)
Democracy (t-1)	0.114**	0.118**	0.116**	0.112**
	(0.034)	(0.035)	(0.034)	(0.035)
Year	0.045*	0.048*	0.047*	0.044*
	(0.021)	(0.023)	(0.023)	(0.022)
Conflict history	0.952*	0.948*	0.931*	0.944*
	(0.413)	(0.411)	(0.414)	(0.417)
Constant	-98.887*	-105.651*	-103.626*	-96.893*
	(43.178)	(47.314)	(47.042)	(44.798)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-250.9	-250.5	-248.7	-248.3

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 11 Logistic regression with interaction effect between Cropland and Prec growth

VARIABLES	(A28) Civil war onset	(A29) Civil war onset	(A30) Civil war onset	(A31) Civil war onset	(A32) Civil war onset
Prec growth (t-1 to t, inv)		0.115	0.161		
		(0.822)	(0.370)		
Cropland x Prec growth (t-1 to t, inv)			0.774		
			(0.478)		
Prec growth (t-2 to t-1, inv)				-1.375	-2.375**
				(0.771)	(0.480)
Cropland x Prec gr (t-2 to t-1, inv)					0.837
					(0.473)
Cropland	0.775	0.775	-0.054	0.773	1.196
	(0.476)	(0.476)	(1.059)	(0.481)	(1.137)
Population (ln)	0.134	0.134	0.134	0.138	0.136
	(0.112)	(0.112)	(0.112)	(0.114)	(0.113)
Political exclusion (t-1)	-1.534	-1.538	-1.538	-1.536	-1.519
	(0.998)	(0.998)	(0.998)	(0.993)	(0.998)
Democracy (t-1)	0.141**	0.141**	0.141**	0.138**	0.139**
	(0.038)	(0.037)	(0.037)	(0.038)	(0.038)
GDP/capita (ln, t-1)	-0.731**	-0.731**	-0.732**	-0.727**	-0.730**
	(0.234)	(0.233)	(0.233)	(0.236)	(0.236)
Year	0.053**	0.053**	0.053**	0.052*	0.052*
	(0.020)	(0.020)	(0.020)	(0.021)	(0.020)
Conflict history	0.739	0.739	0.739	0.760	0.761
	(0.404)	(0.404)	(0.405)	(0.400)	(0.400)
Constant	-104.910**	-105.033**	-105.034**	-103.871**	-104.558**
	(38.538)	(38.386)	(38.393)	(39.129)	(39.106)
Observations	1,251	1,251	1,251	1,251	1,251
Log Likelihood	-247.2	-247.2	-247.2	-245.8	-245.7

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 12 Logistic regression with interaction effect between Cropland and Temp growth

VARIABLES	(A33) Civil war onset	(A34) Civil war onset	(A35) Civil war onset	(A36) Civil war onset
Temp growth (t-1 to t)	-0.943	-0.715		
	(0.930)	(0.726)		
Cropland x Temp growth (t-1 to t)		-0.309		
		(1.429)		
Temp growth (t-2 to t-1)			-0.612	-0.679
			(0.820)	(1.233)
Cropland x Temp growth (t-2 to t-1)				0.092
				(1.588)
Cropland	0.781	0.774	0.778	0.779
	(0.478)	(0.486)	(0.477)	(0.472)
Population (ln)	0.135	0.135	0.134	0.134
	(0.113)	(0.113)	(0.113)	(0.112)
Political exclusion (t-1)	-1.517	-1.520	-1.544	-1.544
	(0.994)	(0.989)	(1.000)	(1.003)
Democracy (t-1)	0.142**	0.142**	0.141**	0.141**
	(0.038)	(0.038)	(0.038)	(0.038)
GDP/capita (ln, t-1)	-0.733**	-0.734**	-0.734**	-0.734**
	(0.234)	(0.234)	(0.234)	(0.234)
Year	0.053**	0.053**	0.053**	0.053**
	(0.020)	(0.020)	(0.020)	(0.020)
Conflict history	0.741	0.742	0.738	0.738
	(0.406)	(0.407)	(0.403)	(0.403)
Constant	-105.243**	-105.252**	-105.684**	-105.687**
	(38.594)	(38.613)	(38.868)	(38.866)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-246.6	-246.5	-246.9	-246.9

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 13 Logistic regression with interaction effect between Cropland and Prec dev

VARIABLES	(A37) Civil war onset	(A38) Civil war onset	(A39) Civil war onset	(A40) Civil war onset
Prec dev (inv)	-0.020	0.319*		
	(0.123)	(0.150)		
Cropland x Prec dev (inv)		-0.386		
		(0.200)		
Prec dev (t-1, inv)			0.074	0.417
			(0.094)	(0.230)
Cropland x Prec dev (t-1, inv)				-0.391
				(0.246)
Cropland	0.778	0.803	0.766	0.842
	(0.477)	(0.492)	(0.483)	(0.521)
Population (ln)	0.134	0.137	0.136	0.140
	(0.113)	(0.113)	(0.112)	(0.112)
Political exclusion (t-1)	-1.530	-1.491	-1.539	-1.464
	(0.989)	(0.977)	(1.018)	(1.008)
Democracy (t-1)	0.141**	0.140**	0.140**	0.139**
	(0.038)	(0.038)	(0.037)	(0.037)
GDP/capita (ln, t-1)	-0.731**	-0.729**	-0.728**	-0.721**
	(0.234)	(0.235)	(0.234)	(0.234)
Year	0.053*	0.053*	0.051*	0.051*
	(0.021)	(0.021)	(0.020)	(0.020)
Conflict history	0.742	0.763	0.734	0.751
	(0.411)	(0.414)	(0.405)	(0.404)
Constant	-105.334**	-105.567**	-102.637**	-102.469**
	(39.386)	(39.490)	(39.092)	(39.013)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-247.1	-246.6	-247.0	-246.4

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 14 Logistic regression with interaction effect between Cropland and Temp dev

VARIABLES	(A41) Civil war onset	(A42) Civil war onset	(A43) Civil war onset	(A44) Civil war onset
Temp dev	-0.238	-0.122		
	(0.140)	(0.143)		
Cropland x Temp dev		-0.134		
		(0.112)		
Temp dev (t-1)			0.004	-0.011
			(0.126)	(0.150)
Cropland x Temp dev (t-1)				0.017
				(0.132)
Cropland	0.802	0.853	0.775	0.766
	(0.474)	(0.454)	(0.481)	(0.480)
Population (ln)	0.125	0.123	0.134	0.134
	(0.117)	(0.118)	(0.114)	(0.114)
Political exclusion (t-1)	-1.508	-1.469	-1.535	-1.542
	(1.054)	(1.055)	(0.988)	(0.980)
Democracy (t-1)	0.143**	0.144**	0.141**	0.141**
	(0.039)	(0.039)	(0.038)	(0.038)
GDP/capita (ln, t-1)	-0.763**	-0.758**	-0.731**	-0.731**
	(0.266)	(0.267)	(0.242)	(0.242)
Year	0.063*	0.063*	0.052*	0.052*
	(0.027)	(0.027)	(0.024)	(0.024)
Conflict history	0.722	0.719	0.740	0.740
	(0.402)	(0.403)	(0.404)	(0.404)
Constant	-125.062*	-124.273*	-104.591*	-104.710*
	(51.332)	(51.360)	(46.266)	(46.299)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-245.1	-245.0	-247.2	-247.2

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 15 Logistic regression with interaction effect between Cropland and Drought

VARIABLES	(A45) Civil war onset	(A46) Civil war onset	(A47) Civil war onset	(A48) Civil war onset
Drought	0.702	1.154		
	(0.638)	(0.767)		
Cropland x Drought		-0.558		
		(1.068)		
Drought (t-1)			1.423*	1.700*
			(0.565)	(0.766)
Cropland x Drought (t-1)				-0.349
				(1.019)
Cropland	0.143	0.140	0.153	0.151
	(0.116)	(0.117)	(0.113)	(0.114)
Population (ln)	-1.465	-1.474	-1.376	-1.376
	(1.014)	(1.017)	(1.028)	(1.029)
Political exclusion (t-1)	0.144**	0.144**	0.146**	0.146**
	(0.035)	(0.036)	(0.036)	(0.036)
Democracy (t-1)	-0.743**	-0.743**	-0.759**	-0.758**
	(0.227)	(0.228)	(0.236)	(0.236)
GDP/capita (ln, t-1)	0.055**	0.055**	0.057**	0.057**
	(0.020)	(0.020)	(0.021)	(0.021)
Year	0.722	0.721	0.706	0.705
	(0.411)	(0.410)	(0.416)	(0.416)
Conflict history	0.758	0.878	0.758	0.849
	(0.474)	(0.462)	(0.465)	(0.600)
Constant	-109.410**	-109.316**	-113.111**	-113.049**
	(37.738)	(37.666)	(40.760)	(40.753)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-246.6	-246.5	-244.4	-244.3

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 16 Logistic regression with interaction effect between Political Exclusion and Prec growth

VARIABLES	(A49) Civil war onset	(A50) Civil war onset	(A51) Civil war onset	(A52) Civil war onset	(A53) Civil war onset
Prec growth (t-1 to t, inv)		0.119	0.015		
		(0.837)	(1.037)		
Exclusion x Prec growth (t-1 to t, inv)			1.112		
			(4.769)		
Prec growth (t-2 to t-1, inv)				-1.409	-0.516
				(0.792)	(0.859)
Exclusion x Prec growth (t-2 to t-1, inv)					-9.205
					(4.841)
Political exclusion (t-1)	-1.163	-1.167	-1.174	-1.153	-1.666
	(0.940)	(0.940)	(0.940)	(0.938)	(0.959)
Irrigated cropland	-0.006	-0.006	-0.006	-0.006	-0.007
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
Rainfed cropland	0.008	0.008	0.008	0.008	0.008
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Population (ln)	0.235	0.235	0.235	0.240	0.236
	(0.125)	(0.125)	(0.125)	(0.126)	(0.124)
Democracy (t-1)	0.141**	0.142**	0.142**	0.139**	0.139**
	(0.040)	(0.039)	(0.039)	(0.040)	(0.040)
GDP/capita (ln, t-1)	-0.748**	-0.748**	-0.748**	-0.748**	-0.760**
	(0.279)	(0.278)	(0.278)	(0.280)	(0.277)
Year	0.053*	0.053*	0.053*	0.053*	0.054*
	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)
Conflict history	0.812*	0.812*	0.812*	0.830*	0.838*
	(0.405)	(0.405)	(0.404)	(0.400)	(0.402)
Constant	-106.459*	-106.573*	-106.528*	-106.015*	-108.589*
	(43.356)	(43.223)	(43.209)	(43.583)	(43.241)
Observations	1,251	1,251	1,251	1,251	1,251
Log Likelihood	-248.7	-248.7	-248.7	-247.4	-246.7

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 17 Logistic regression with interaction effect between Exclusion and Temp growth

VARIABLES	(A54) Civil war onset	(A55) Civil war onset	(A56) Civil war onset	(A57) Civil war onset
Temp growth (t-1 to t)	-0.911	-1.016		
	(0.897)	(1.154)		
Exclusion x Temp growth (t-1 to t)		1.014		
		(4.386)		
Temp growth (t-2 to t-1)			-0.590	-0.769
			(0.807)	(0.904)
Exclusion x Temp growth (t-2 to t-1)				1.738
				(4.909)
Political exclusion (t-1)	-1.140	-1.132	-1.171	-1.162
	(0.940)	(0.944)	(0.941)	(0.940)
Irrigated cropland	-0.006	-0.006	-0.006	-0.006
	(0.012)	(0.012)	(0.012)	(0.012)
Rainfed cropland	0.008	0.008	0.008	0.008
	(0.015)	(0.015)	(0.015)	(0.015)
Population (ln)	0.237	0.237	0.235	0.235
	(0.125)	(0.125)	(0.126)	(0.126)
Democracy (t-1)	0.142**	0.142**	0.142**	0.142**
	(0.040)	(0.040)	(0.040)	(0.040)
GDP/capita (ln, t-1)	-0.750**	-0.749**	-0.751**	-0.751**
	(0.279)	(0.279)	(0.280)	(0.280)
Year	0.053*	0.053*	0.054*	0.053*
	(0.023)	(0.023)	(0.023)	(0.023)
Conflict history	0.814*	0.814*	0.811*	0.811*
	(0.406)	(0.406)	(0.404)	(0.404)
Constant	-106.673*	-106.543*	-107.179*	-107.152*
	(43.371)	(43.276)	(43.667)	(43.628)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-248.2	-248.2	-248.5	-248.5

Robust standard errors in parentheses;** p<0.01, * p<0.05

Table 18 Logistic regression with interaction effect between Exclusion and Prec dev

VARIABLES	(A58) Civil war onset	(A59) Civil war onset	(A60) Civil war onset	(A61) Civil war onset
Prec dev (inv)	-0.001	-0.022		
	(0.120)	(0.143)		
Exclusion x Prec dev (inv)		0.207		
		(0.689)		
Prec dev (t-1, inv)			0.091	0.078
			(0.089)	(0.098)
Exclusion x Prec dev (t-1, inv)				0.130
				(0.629)
Political exclusion (t-1)	-1.163	-1.161	-1.163	-1.165
	(0.939)	(0.964)	(0.959)	(0.981)
Irrigated cropland	-0.006	-0.006	-0.007	-0.007
	(0.012)	(0.012)	(0.012)	(0.012)
Rainfed cropland	0.008	0.008	0.008	0.008
	(0.015)	(0.015)	(0.015)	(0.015)
Population (ln)	0.235	0.235	0.239	0.239
	(0.127)	(0.127)	(0.122)	(0.122)
Democracy (t-1)	0.141**	0.141**	0.142**	0.141**
	(0.039)	(0.039)	(0.039)	(0.039)
GDP/capita (ln, t-1)	-0.748**	-0.746**	-0.748**	-0.747**
	(0.278)	(0.279)	(0.280)	(0.281)
Year	0.053*	0.053*	0.052*	0.052*
	(0.023)	(0.023)	(0.023)	(0.023)
Conflict history	0.812*	0.812*	0.805*	0.805*
	(0.410)	(0.410)	(0.407)	(0.407)
Constant	-106.473*	-105.859*	-104.346*	-104.022*
	(43.913)	(44.155)	(43.985)	(44.418)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-248.7	-248.7	-248.4	-248.4

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 19 Logistic regression with interaction effect between Exclusion and Temp dev

VARIABLES	(A62) Civil war onset	(A63) Civil war onset	(A64) Civil war onset	(A65) Civil war onset
Temp dev	-0.242	-0.219		
	(0.143)	(0.172)		
Exclusion x Temp dev		-0.221		
		(0.452)		
Temp dev (t-1)			0.005	0.058
			(0.128)	(0.116)
Exclusion x Temp dev (t-1)				-0.513
				(0.686)
Political exclusion (t-1)	-1.126	-1.067	-1.164	-0.938
	(0.981)	(0.972)	(0.937)	(0.907)
Irrigated cropland	-0.008	-0.008	-0.006	-0.006
	(0.013)	(0.013)	(0.012)	(0.012)
Rainfed cropland	0.008	0.008	0.008	0.008
	(0.015)	(0.015)	(0.015)	(0.015)
Population (ln)	0.237	0.237	0.235	0.237
	(0.127)	(0.126)	(0.125)	(0.125)
Democracy (t-1)	0.145**	0.144**	0.141**	0.141**
	(0.042)	(0.042)	(0.040)	(0.040)
GDP/capita (ln, t-1)	-0.790*	-0.792*	-0.747*	-0.749*
	(0.317)	(0.316)	(0.292)	(0.292)
Year	0.064*	0.064*	0.053	0.053
	(0.030)	(0.030)	(0.027)	(0.027)
Conflict history	0.801*	0.804*	0.812*	0.820*
	(0.404)	(0.406)	(0.404)	(0.406)
Constant	-128.245*	-128.044*	-106.051*	-105.494*
	(56.669)	(56.743)	(51.884)	(51.636)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-246.7	-246.6	-248.7	-248.6

Robust standard errors in parentheses;** p<0.01, * p<0.05

Table 20 Logistic regression with interaction effect between Exclusion and Drought

VARIABLES	(A66) Civil war onset	(A67) Civil war onset	(A68) Civil war onset	(A69) Civil war onset
Drought	0.734	0.882		
	(0.605)	(0.668)		
Exclusion x Drought		-1.446		
		(2.037)		
Drought (t-1)			1.428*	1.100
			(0.563)	(0.706)
Exclusion x Drought (t-1)				2.868
				(1.837)
Political exclusion (t-1)	-1.090	-0.830	-0.985	-1.737
	(0.957)	(0.969)	(0.987)	(1.131)
Irrigated cropland	-0.005	-0.005	-0.005	-0.005
	(0.012)	(0.012)	(0.013)	(0.012)
Rainfed cropland	0.008	0.009	0.009	0.009
	(0.015)	(0.015)	(0.015)	(0.016)
Population (ln)	0.238	0.237	0.242	0.242
	(0.126)	(0.126)	(0.124)	(0.124)
Democracy (t-1)	0.144**	0.144**	0.145**	0.145**
	(0.038)	(0.037)	(0.038)	(0.038)
GDP/capita (ln, t-1)	-0.760**	-0.763**	-0.773**	-0.775**
	(0.274)	(0.273)	(0.284)	(0.284)
Year	0.055*	0.056*	0.057*	0.057*
	(0.022)	(0.022)	(0.024)	(0.024)
Conflict history	0.785	0.782	0.764	0.774
	(0.412)	(0.411)	(0.415)	(0.417)
Constant	-111.212**	-111.754**	-114.491*	-114.539*
	(42.671)	(42.533)	(45.823)	(45.714)
Observations	1,251	1,251	1,251	1,251
Log Likelihood	-248.1	-248.0	-245.9	-245.6

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 21 Fixed-effects OLS

VARIABLES	(A70) Civil war onset
Prec	0.004
	(0.034)
Prec (t-1)	-0.027
	(0.034)
Temp	-0.014
	(0.014)
Temp (t-1)	0.009
	(0.014)
Constant	0.179
	(0.257)
Observations	1,367
Number of Countries	30
R-squared	0.001

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 22 OLS - GDP and rainfall growth – First Stage

VARIABLES	(A71) GDP growth	(A72) GDP growth	(A73) GDP growth
Rain growth	0.010	0.012	0.014
	(0.014)	(0.014)	(0.014)
Rain growth (t-1)	0.014	0.015	0.018
	(0.013)	(0.013)	(0.014)
Exclusion (t-1)		0.014	0.002
		(0.030)	(0.037)
Population (t-1)		0.005	0.025*
		(0.003)	(0.009)
Democracy (t-1)		-0.001	-0.003**
		(0.001)	(0.001)
Constant	0.066**	0.017	-0.206*
	(0.004)	(0.028)	(0.079)
Observations	1,313	1,250	1,250
R-squared	0.001	0.012	0.105

Robust standard errors in parentheses; ** p<0.01, * p<0.05
Model A73 includes country fixed effects

Table 23 OLS - GDP and temperature growth – First Stage

VARIABLES	(A74) GDP growth	(A75) GDP growth	(A76) GDP growth
Temp growth	-0.007	-0.001	>-0.001
	(0.012)	(0.012)	(0.012)
Temp growth (t-1)	-0.002	0.005	0.007
	(0.011)	(0.010)	(0.009)
Exclusion (t-1)		0.013	0.001
		(0.030)	(0.038)
Population (t-1)		0.005	0.025*
		(0.003)	(0.009)
Democracy (t-1)		-0.001	-0.003**
		(0.001)	(0.001)
Constant	0.066**	0.018	-0.204*
	(0.004)	(0.028)	(0.079)
Observations	1,313	1,250	1,250
R-squared	<0.001	0.011	0.104

Robust standard errors in parentheses; ** p<0.01, * p<0.05
Model A76 includes country fixed effects

Table 24 OLS Food production and rainfall growth – First Stage

VARIABLES	(A77) Food growth	(A78) Food growth	(A79) Food growth
Rain growth	0.035	0.043	0.042
	(0.022)	(0.022)	(0.022)
Rain growth (t-1)	0.008	0.005	0.004
	(0.020)	(0.017)	(0.017)
Exclusion (t-1)		-0.012	0.008
		(0.014)	(0.034)
Population (t-1)		0.002	0.014*
		(0.002)	(0.005)
Democracy (t-1)		>-0.001	0.001
		(0.001)	(0.001)
Constant	0.010**	-0.004	-0.166*
	(0.003)	(0.022)	(0.074)
Observations	1,079	1,014	1,014
R-squared	0.004	0.011	0.043

Robust standard errors in parentheses; ** p<0.01, * p<0.05
Model A79 includes country fixed effects

Table 25 OLS Food production and temperature growth – First Stage

VARIABLES	(A80) Food growth	(A81) Food growth	(A82) Food growth
Temp growth	-0.007	-0.004	-0.004
	(0.016)	(0.016)	(0.016)
Temp growth (t-1)	0.003	>-0.001	<0.001
	(0.016)	(0.015)	(0.016)
Exclusion (t-1)		-0.013	0.002
		(0.014)	(0.034)
Population (t-1)		0.002	0.014*
		(0.002)	(0.005)
Democracy (t-1)		<0.001	0.001
		(0.001)	(0.001)
Constant	0.011**	-0.003	-0.172*
	(0.003)	(0.022)	(0.074)
Observations	1,079	1,014	1,014
R-squared	<0.001	0.004	0.036

Robust standard errors in parentheses; ** p<0.01, * p<0.05
Model A82 includes country fixed effects

Table 26 Instrumental Variables 2SLS

VARIABLES	(A83) Civil war onset	(A84) Civil war onset	(A85) Civil war onset	(A86) Civil war onset
GDP growth	4.591	4.217	1.870	3.058
	(3.783)	(3.527)	(4.503)	(5.630)
GDP growth (t-1)	1.539	1.595	2.723	2.081
	(5.472)	(5.286)	(3.644)	(3.763)
Conflict history	0.064	-0.020	0.032	-0.038
	(0.093)	(0.099)	(0.099)	(0.134)
Constant	0.993	1.037	0.183	0.959
	(4.188)	(0.610)	(3.229)	(0.692)
Observations	1,293	1,293	1,293	1,293

Robust standard errors in parentheses; ** p<0.01, * p<0.05
GDP growth is instrumented by rainfall growth (Models A83 and A84) or temperature growth (Models A85 and A86); Models A84 and A86 include country fixed effects

Table 27 Instrumental Variables 2SLS

VARIABLES	(A87) Civil war onset	(A88) Civil war onset	(A89) Civil war onset	(A90) Civil war onset
Food growth	4.616	5.226	0.306	5.196
	(6.609)	(8.257)	(8.204)	(22.796)
Food growth (t-1)	4.388	4.651	4.438	13.786
	(4.886)	(5.769)	(6.538)	(41.427)
Conflict history	0.054	-0.052	-0.009	-0.007
	(0.157)	(0.163)	(0.094)	(0.468)
Constant	3.630	1.360	0.607	2.437
	(7.566)	(1.292)	(3.930)	(5.619)
Observations	1,055	1,055	1,055	1,055

Robust standard errors in parentheses; ** p<0.01, * p<0.05
Food growth is instrumented by rainfall growth (Models A87 and A88) or temperature growth (Models A89 and A90); Models A88 and A90 include country fixed effects

Table 28 Civil war onset limited to 1000+ battle-related deaths

VARIABLES	(A91) Civil war onset	(A92) Civil war onset	(A93) Civil war onset	(A94) Civil war onset	(A95) Civil war onset	(A96) Civil war onset
Precipitation					-0.006	-0.007
					(0.026)	(0.027)
Temperature					-0.018	-0.018
					(0.026)	(0.027)
Precipitation (t-1)					-0.018	-0.009
					(0.011)	(0.012)
Temperature (t-1)					0.003	0.012
					(0.011)	(0.012)
Prec growth (t-1 to t)		0.228				
		(1.235)				
Prec growth (t-2 to t-1)		1.550				
		(1.173)				
Temp growth (t-1 to t)		-1.545				
		(1.160)				
Temp growth (t-2 to t-1)		-0.700				
		(1.192)				
Prec dev			-0.075			
			(0.162)			
Prec dev (t-1)			-0.168			
			(0.162)			
Temp dev			-0.306			
			(0.163)			
Temp dev (t-1)			0.055			
			(0.161)			
Drought				0.716		
				(0.771)		
Drought (t-1)				1.979**		
				(0.683)		
Irrigated cropland	-0.035*	-0.035*	-0.038*	-0.032*		
	(0.016)	(0.015)	(0.016)	(0.015)		
Rainfed cropland	-0.025	-0.024	-0.023	-0.024		
	(0.026)	(0.026)	(0.026)	(0.027)		
Population (ln)	0.267*	0.269*	0.267*	0.281*		
	(0.115)	(0.116)	(0.115)	(0.117)		
Political exclusion (t-1)	0.661	0.722	0.766	0.928		
	(1.253)	(1.257)	(1.290)	(1.239)		
Democracy (t-1)	0.174**	0.173**	0.178**	0.184**		
	(0.058)	(0.058)	(0.058)	(0.060)		
GDP/capita (ln, t-1)	-0.935**	-0.952**	-0.990**	-0.996**		
	(0.268)	(0.270)	(0.274)	(0.267)		
Year	0.060**	0.061**	0.069**	0.068**	0.001	
	(0.020)	(0.020)	(0.023)	(0.021)	(<0.001)	
Conflict history	0.572	0.577	0.552	0.472		
	(0.393)	(0.395)	(0.390)	(0.400)		
Constant	-119.304**	-121.843**	-136.779**	-135.479**	-0.700	0.006
	(38.529)	(38.720)	(43.239)	(40.066)	(0.616)	(0.273)
Observations	1,251	1,251	1,251	1,251	1,367	1,367
R-Squared					0.004	0.032
Log Likelihood	-166.4	-164.4	-163.9	-162.0		
Number of Countries					30	30

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 29 Top 10 countries at risk of civil war onset at end of sample period

Controls only		Model 2*a*	
Country	Pr	Country	Pr
India	0.31	India	0.27
Myanmar	0.22	Philippines	0.25
Philippines	0.21	Indonesia	0.22
Nepal	0.19	Nepal	0.18
Afghanistan	0.17	Myanmar	0.17
Indonesia	0.13	Sri Lanka	0.14
Sri Lanka	0.10	Afghanistan	0.13
Pakistan	0.07	Thailand	0.11
Thailand	0.07	Pakistan	0.07
Bangladesh	0.06	Vietnam	0.05

V. Grid-level analysis

Table 30 Rare events logistic regression

VARIABLES	(A97) Civil war onset	(A98) Civil war onset	(A99) Civil war onset	(A100) Civil war onset	(A101) Civil war onset	(A102) Civil war onset
Precipitation		0.182				
		(0.147)				
Temperature		0.073				
		(0.040)				
Precipitation (t-1)			0.264			
			(0.137)			
Temperature (t-1)			0.070			
			(0.040)			
Prec growth (t-1 to t)				-0.773		
				(0.533)		
Prec growth (t-2 to t-1)				0.174		
				(0.492)		
Temp growth (t-1 to t)				-1.213*		
				(0.492)		
Temp growth (t-2 to t-1)				-0.738		
				(0.474)		
Prec dev					-0.283	
					(0.166)	
Prec dev (t-1)					0.038	
					(0.089)	
Temp dev					-0.255**	
					(0.093)	
Temp dev (t-1)					0.050	
					(0.171)	
Drought						-0.116
						(0.244)
Drought (t-1)						0.362*
						(0.183)
Border distance (ln)	-1.622*	-1.562*	-1.545*	-1.619*	-1.613*	-1.622*
	(0.698)	(0.648)	(0.651)	(0.688)	(0.650)	(0.700)
Capital distance (ln)	-1.122	-1.460	-1.437	-1.099	-0.965	-1.108
	(2.117)	(2.016)	(1.990)	(2.134)	(2.154)	(2.123)
Irrigated cropland (>10%)	-0.800*	-0.718	-0.670	-0.802*	-0.829*	-0.792*
	(0.353)	(0.381)	(0.385)	(0.353)	(0.363)	(0.354)
Rainfed cropland (>10%)	-0.641	-0.640	-0.642	-0.645	-0.640	-0.641
	(0.353)	(0.328)	(0.328)	(0.355)	(0.353)	(0.355)
Local population (ln)	0.421**	0.299**	0.298**	0.423**	0.422**	0.421**
	(0.100)	(0.111)	(0.108)	(0.102)	(0.101)	(0.101)
Local political exclusion (t-1)	1.493**	1.509**	1.493**	1.491**	1.496**	1.490**
	(0.540)	(0.524)	(0.516)	(0.540)	(0.544)	(0.539)
Democracy (t-1)	0.155*	0.113*	0.113*	0.156*	0.156*	0.156*
	(0.062)	(0.055)	(0.054)	(0.064)	(0.061)	(0.061)
GDP/capita (ln, t-1)	-0.391	-0.487	-0.489	-0.397	-0.376	-0.391
	(0.366)	(0.306)	(0.294)	(0.363)	(0.357)	(0.366)
Year	0.030	0.040	0.041	0.031	0.033	0.030
	(0.032)	(0.026)	(0.024)	(0.031)	(0.034)	(0.032)
Local conflict history	1.354*	0.832*	0.811*	1.353*	1.379*	1.350*
	(0.585)	(0.370)	(0.363)	(0.583)	(0.566)	(0.590)
Constant	-69.728	-88.443	-89.851	-71.628	-76.632	-70.608
	(59.932)	(48.191)	(46.129)	(59.387)	(64.912)	(60.296)
Observations	521,258	521,258	521,258	521,258	521,258	521,258
Log Likelihood						

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 31 Logistic regression with interaction effect between Cropland and Prec growth

VARIABLES	(A103) Civil war onset	(A104) Civil war onset	(A105) Civil war onset	(A106) Civil war onset	(A107) Civil war onset
Prec growth (t-1 to t, inv)		0.941	2.010**		
		(0.524)	(0.518)		
Cropland x Prec growth (t-1 to t, inv)			-1.172		
			(0.634)		
Prec growth (t-2 to t-1, inv)				-0.532	-0.051
				(0.335)	(1.010)
Cropland x Prec growth (t-2 to t-1, inv)					-0.507
					(0.954)
Cropland	1.149**	1.150**	1.203**	1.150**	1.122**
	(0.338)	(0.329)	(0.313)	(0.341)	(0.315)
Border distance (ln)	-1.481*	-1.532*	-1.523*	-1.484*	-1.486*
	(0.727)	(0.695)	(0.694)	(0.719)	(0.721)
Capital distance (ln)	-0.123	-0.367	-0.365	-0.186	-0.183
	(2.140)	(2.138)	(2.132)	(2.147)	(2.150)
Irrigated cropland (>10%)	-0.972*	-0.964*	-0.964*	-0.968*	-0.968*
	(0.390)	(0.388)	(0.389)	(0.384)	(0.383)
Rainfed cropland (>10%)	-0.911**	-0.892**	-0.896**	-0.909**	-0.907**
	(0.281)	(0.276)	(0.277)	(0.277)	(0.277)
Local population (ln)	0.432**	0.436**	0.435**	0.433**	0.432**
	(0.099)	(0.100)	(0.101)	(0.098)	(0.098)
Local political exclusion (t-1)	1.380*	1.387*	1.381*	1.384*	1.384*
	(0.612)	(0.606)	(0.599)	(0.612)	(0.614)
Democracy (t-1)	0.194*	0.190*	0.188*	0.188*	0.189*
	(0.083)	(0.082)	(0.082)	(0.084)	(0.085)
GDP/capita (ln, t-1)	-0.490	-0.506	-0.507	-0.492	-0.492
	(0.279)	(0.278)	(0.280)	(0.285)	(0.284)
Year	0.044*	0.044*	0.045*	0.044*	0.044*
	(0.021)	(0.020)	(0.021)	(0.021)	(0.021)
Local conflict history	3.247**	3.279**	3.278**	3.287**	3.288**
	(0.524)	(0.523)	(0.524)	(0.528)	(0.528)
Constant	-95.161*	-96.298*	-96.997*	-95.565*	-95.458*
	(38.912)	(38.673)	(38.846)	(39.853)	(39.749)
Observations	23,356	23,356	23,356	23,356	23,356
Log Likelihood	-395.6	-393.8	-393.6	-394.4	-394.4

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 32 Logistic regression with interaction effect between Cropland and Temp growth

VARIABLES	(A108) Civil war onset	(A109) Civil war onset	(A110) Civil war onset	(A111) Civil war onset
Temp growth (t-1 to t)	-0.522	-1.790**		
	(0.618)	(0.672)		
Cropland x Temp growth (t-1 to t)		1.489		
		(0.982)		
Temp growth (t-2 to t-1)			-0.584	1.037
			(0.565)	(1.160)
Cropland x Temp growth (t-2 to t-1)				-1.835
				(1.257)
Cropland	1.147**	1.218**	1.151**	1.166**
	(0.340)	(0.340)	(0.337)	(0.370)
Border distance (ln)	-1.485*	-1.491*	-1.461*	-1.471*
	(0.724)	(0.726)	(0.726)	(0.732)
Capital distance (ln)	-0.119	-0.122	-0.101	-0.092
	(2.158)	(2.153)	(2.116)	(2.115)
Irrigated cropland (>10%)	-0.983*	-0.986*	-0.952*	-0.946*
	(0.385)	(0.386)	(0.385)	(0.386)
Rainfed cropland (>10%)	-0.914**	-0.916**	-0.910**	-0.910**
	(0.279)	(0.281)	(0.280)	(0.280)
Local population (ln)	0.433**	0.436**	0.431**	0.433**
	(0.100)	(0.101)	(0.098)	(0.098)
Local political exclusion (t-1)	1.376*	1.382*	1.380*	1.383*
	(0.611)	(0.617)	(0.615)	(0.618)
Democracy (t-1)	0.196*	0.196*	0.194*	0.194*
	(0.084)	(0.084)	(0.083)	(0.084)
GDP/capita (ln, t-1)	-0.497	-0.497	-0.488	-0.489
	(0.276)	(0.276)	(0.279)	(0.280)
Year	0.044*	0.044*	0.044*	0.044*
	(0.020)	(0.020)	(0.021)	(0.021)
Local conflict history	3.233**	3.242**	3.267**	3.278**
	(0.524)	(0.521)	(0.519)	(0.519)
Constant	-95.988*	-96.171*	-95.808*	-95.671*
	(38.653)	(38.733)	(38.815)	(38.918)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-395.3	-394.9	-395.2	-394.7

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 33 Logistic regression with interaction effect between Cropland and Prec dev

VARIABLES	(A112) Civil war onset	(A113) Civil war onset	(A114) Civil war onset	(A115) Civil war onset
Prec dev (inv)	0.177	0.373		
	(0.158)	(0.243)		
Cropland x Prec dev (inv)		-0.214		
		(0.228)		
Prec dev (t-1, inv)			-0.083	-0.209
			(0.089)	(0.299)
Cropland x Prec dev (t-1, inv)				0.141
				(0.315)
Cropland	1.126**	1.176**	1.159**	1.177**
	(0.337)	(0.346)	(0.337)	(0.314)
Border distance (ln)	-1.531*	-1.523*	-1.480*	-1.483*
	(0.691)	(0.693)	(0.732)	(0.737)
Capital distance (ln)	-0.175	-0.161	-0.200	-0.215
	(2.112)	(2.105)	(2.174)	(2.161)
Irrigated cropland (>10%)	-0.947*	-0.947*	-0.979*	-0.980*
	(0.401)	(0.400)	(0.390)	(0.391)
Rainfed cropland (>10%)	-0.889**	-0.891**	-0.912**	-0.912**
	(0.281)	(0.281)	(0.280)	(0.281)
Local population (ln)	0.434**	0.434**	0.433**	0.433**
	(0.101)	(0.101)	(0.099)	(0.100)
Local political exclusion (t-1)	1.370*	1.368*	1.388*	1.385*
	(0.603)	(0.601)	(0.614)	(0.609)
Democracy (t-1)	0.188*	0.188*	0.195*	0.194*
	(0.082)	(0.082)	(0.082)	(0.082)
GDP/capita (ln, t-1)	-0.475	-0.476	-0.504	-0.504
	(0.284)	(0.285)	(0.280)	(0.280)
Year	0.041	0.041	0.045*	0.045*
	(0.022)	(0.022)	(0.021)	(0.021)
Local conflict history	3.253**	3.251**	3.257**	3.259**
	(0.511)	(0.511)	(0.527)	(0.527)
Constant	-89.399*	-89.959*	-98.266*	-98.503*
	(40.592)	(40.801)	(39.193)	(39.180)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-394.5	-394.4	-395.4	-395.3

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 34 Logistic regression with interaction effect between Cropland and Temp dev

VARIABLES	(A116) Civil war onset	(A117) Civil war onset	(A118) Civil war onset	(A119) Civil war onset
Temp dev	-0.179	-0.233*		
	(0.109)	(0.101)		
Cropland x Temp dev		0.061		
		(0.160)		
Temp dev (t-1)			-0.059	0.233*
			(0.152)	(0.109)
Cropland x Temp dev (t-1)				-0.321
				(0.229)
Cropland	1.144**	1.134**	1.150**	1.306**
	(0.340)	(0.351)	(0.337)	(0.359)
Border distance (ln)	-1.444*	-1.442*	-1.467*	-1.472*
	(0.722)	(0.721)	(0.726)	(0.729)
Capital distance (ln)	-0.050	-0.050	-0.115	-0.131
	(2.109)	(2.109)	(2.115)	(2.119)
Irrigated cropland (>10%)	-0.965*	-0.967*	-0.968*	-0.964*
	(0.385)	(0.384)	(0.381)	(0.378)
Rainfed cropland (>10%)	-0.906**	-0.906**	-0.906**	-0.909**
	(0.283)	(0.283)	(0.279)	(0.280)
Local population (ln)	0.428**	0.428**	0.430**	0.431**
	(0.099)	(0.099)	(0.099)	(0.100)
Local political exclusion (t-1)	1.376*	1.375*	1.381*	1.393*
	(0.618)	(0.618)	(0.614)	(0.614)
Democracy (t-1)	0.196*	0.196*	0.193*	0.194*
	(0.085)	(0.085)	(0.083)	(0.083)
GDP/capita (ln, t-1)	-0.488	-0.487	-0.487	-0.486
	(0.274)	(0.274)	(0.272)	(0.272)
Year	0.048*	0.048*	0.045*	0.044*
	(0.021)	(0.021)	(0.022)	(0.022)
Local conflict history	3.252**	3.252**	3.257**	3.260**
	(0.517)	(0.516)	(0.529)	(0.530)
Constant	-103.513**	-103.423**	-97.501*	-97.046*
	(39.587)	(39.622)	(41.881)	(41.979)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-394.2	-394.2	-395.5	-395.0

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 35 Logistic regression with interaction effect between Cropland and Drought

VARIABLES	(A120) Civil war onset	(A121) Civil war onset	(A122) Civil war onset	(A123) Civil war onset
Drought	-0.078	0.704		
	(0.304)	(0.663)		
Cropland x Drought		-0.891		
		(0.662)		
Drought (t-1)			0.321	0.761
			(0.249)	(0.730)
Cropland x Drought (t-1)				-0.490
				(0.804)
Cropland	1.152**	1.323**	1.140**	1.247**
	(0.342)	(0.389)	(0.343)	(0.447)
Border distance (ln)	-1.480*	-1.478*	-1.452*	-1.463*
	(0.727)	(0.728)	(0.735)	(0.737)
Capital distance (ln)	-0.132	-0.161	0.012	-0.013
	(2.161)	(2.154)	(2.121)	(2.148)
Irrigated cropland (>10%)	-0.977*	-0.988*	-0.968*	-0.971*
	(0.398)	(0.398)	(0.394)	(0.396)
Rainfed cropland (>10%)	-0.911**	-0.913**	-0.917**	-0.916**
	(0.279)	(0.279)	(0.287)	(0.286)
Local population (ln)	0.430**	0.431**	0.430**	0.431**
	(0.100)	(0.101)	(0.099)	(0.099)
Local political exclusion (t-1)	1.378*	1.388*	1.387*	1.392*
	(0.615)	(0.619)	(0.609)	(0.612)
Democracy (t-1)	0.193*	0.194*	0.198*	0.199*
	(0.083)	(0.084)	(0.083)	(0.084)
GDP/capita (ln, t-1)	-0.488	-0.486	-0.491	-0.486
	(0.278)	(0.278)	(0.279)	(0.276)
Year	0.043*	0.043*	0.045*	0.044*
	(0.021)	(0.021)	(0.021)	(0.021)
Local conflict history	3.247**	3.247**	3.234**	3.239**
	(0.525)	(0.525)	(0.524)	(0.527)
Constant	-94.490*	-94.359*	-97.448*	-96.567*
	(38.978)	(39.033)	(39.848)	(39.320)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-395.6	-395.2	-395.0	-394.9

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 36 Logistic regression with interaction effect between local political exclusion and Prec growth

VARIABLES	(A124) Civil war onset	(A125) Civil war onset	(A126) Civil war onset	(A127) Civil war onset	(A128) Civil war onset
Prec growth (t-1 to t, inv)		0.943	1.021		
		(0.539)	(0.959)		
Exclusion x Prec growth (t-1 to t, inv)			-0.124		
			(0.974)		
Prec growth (t-2 to t-1, inv)				-0.547	-0.572
				(0.371)	(0.539)
Exclusion x Prec growth (t-2 to t, inv)					0.042
					(0.389)
Local political exclusion (t-1)	1.357*	1.361*	1.363*	1.363*	1.367*
	(0.597)	(0.587)	(0.578)	(0.595)	(0.587)
Border distance (ln)	-1.461*	-1.505*	-1.504*	-1.465*	-1.465*
	(0.741)	(0.712)	(0.714)	(0.734)	(0.734)
Capital distance (ln)	-0.037	-0.272	-0.260	-0.091	-0.089
	(2.102)	(2.116)	(2.086)	(2.109)	(2.107)
Irrigated cropland (>10%)	-0.797*	-0.788*	-0.787*	-0.794*	-0.794*
	(0.375)	(0.375)	(0.373)	(0.370)	(0.369)
Rainfed cropland (>10%)	-0.709*	-0.692*	-0.691*	-0.708**	-0.708**
	(0.276)	(0.270)	(0.271)	(0.273)	(0.273)
Local population (ln)	0.466**	0.469**	0.469**	0.468**	0.468**
	(0.093)	(0.094)	(0.094)	(0.092)	(0.092)
Democracy (t-1)	0.185*	0.180*	0.180*	0.179*	0.179*
	(0.075)	(0.075)	(0.076)	(0.077)	(0.077)
GDP/capita (ln, t-1)	-0.460	-0.474	-0.474	-0.463	-0.462
	(0.279)	(0.280)	(0.277)	(0.285)	(0.284)
Year	0.041*	0.041*	0.041*	0.041	0.041
	(0.021)	(0.021)	(0.020)	(0.021)	(0.021)
Local conflict history	3.258**	3.296**	3.297**	3.299**	3.299**
	(0.558)	(0.559)	(0.560)	(0.562)	(0.562)
Constant	-88.948*	-90.069*	-89.971*	-89.384*	-89.291*
	(39.149)	(38.925)	(38.706)	(40.050)	(39.773)
Observations	23,356	23,356	23,356	23,356	23,356
Log Likelihood	-400.5	-398.6	-398.6	-399.3	-399.3

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 37 Logistic regression with interaction effect between local political exclusion and Temp growth

VARIABLES	(A129) Civil war onset	(A130) Civil war onset	(A131) Civil war onset	(A132) Civil war onset
Temp growth (t-1 to t)	-0.524	-1.433		
	(0.615)	(1.048)		
Exclusion x Temp growth (t-1 to t)		1.236		
		(1.096)		
Temp growth (t-2 to t-1)			-0.569	-0.892
			(0.559)	(0.952)
Exclusion x Temp growth (t-2 to t-1)				0.445
				(1.070)
Local political exclusion (t-1)	1.353*	1.387*	1.357*	1.363*
	(0.596)	(0.613)	(0.599)	(0.599)
Border distance (ln)	-1.467*	-1.454*	-1.440	-1.445*
	(0.737)	(0.732)	(0.739)	(0.736)
Capital distance (ln)	-0.030	0.001	-0.016	-0.026
	(2.120)	(2.133)	(2.079)	(2.070)
Irrigated cropland (>10%)	-0.808*	-0.797*	-0.778*	-0.774*
	(0.370)	(0.370)	(0.372)	(0.372)
Rainfed cropland (>10%)	-0.712**	-0.711**	-0.708**	-0.708**
	(0.275)	(0.275)	(0.275)	(0.275)
Local population (ln)	0.468**	0.468**	0.465**	0.465**
	(0.094)	(0.094)	(0.092)	(0.092)
Democracy (t-1)	0.187*	0.186*	0.185*	0.185*
	(0.076)	(0.076)	(0.076)	(0.076)
GDP/capita (ln, t-1)	-0.466	-0.460	-0.458	-0.459
	(0.276)	(0.274)	(0.280)	(0.279)
Year	0.041*	0.041*	0.041*	0.041*
	(0.020)	(0.020)	(0.021)	(0.021)
Local conflict history	3.244**	3.257**	3.277**	3.280**
	(0.558)	(0.553)	(0.553)	(0.554)
Constant	-89.818*	-88.902*	-89.590*	-89.467*
	(38.768)	(38.490)	(39.138)	(39.029)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-400.2	-399.8	-400.1	-400.0

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 38 Logistic regression with interaction effect between local political exclusion and Prec dev

VARIABLES	(A133) Civil war onset	(A134) Civil war onset	(A135) Civil war onset	(A136) Civil war onset
Prec dev (inv)	0.197	0.088		
	(0.153)	(0.140)		
Exclusion x Prec dev (inv)		0.153		
		(0.210)		
Prec dev (t-1, inv)			-0.068	-0.256
			(0.096)	(0.176)
Exclusion x Prec dev (t-1, inv)				0.272
				(0.180)
Local political exclusion (t-1)	1.347*	1.325*	1.363*	1.383*
	(0.586)	(0.581)	(0.596)	(0.581)
Border distance (ln)	-1.507*	-1.516*	-1.462*	-1.481*
	(0.708)	(0.709)	(0.744)	(0.747)
Capital distance (ln)	-0.095	-0.117	-0.099	-0.054
	(2.085)	(2.072)	(2.141)	(2.125)
Irrigated cropland (>10%)	-0.773*	-0.769*	-0.801*	-0.781*
	(0.384)	(0.387)	(0.375)	(0.369)
Rainfed cropland (>10%)	-0.691*	-0.693*	-0.708**	-0.708*
	(0.274)	(0.275)	(0.274)	(0.276)
Local population (ln)	0.467**	0.467**	0.467**	0.465**
	(0.094)	(0.095)	(0.094)	(0.093)
Democracy (t-1)	0.179*	0.178*	0.185*	0.184*
	(0.074)	(0.075)	(0.075)	(0.075)
GDP/capita (ln, t-1)	-0.445	-0.449	-0.470	-0.468
	(0.284)	(0.282)	(0.281)	(0.281)
Year	0.038	0.038	0.042*	0.041*
	(0.021)	(0.021)	(0.021)	(0.021)
Local conflict history	3.267**	3.263**	3.268**	3.262**
	(0.544)	(0.544)	(0.562)	(0.557)
Constant	-83.341*	-83.454*	-91.353*	-90.591*
	(40.337)	(40.268)	(39.370)	(39.379)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-399.1	-399.0	-400.3	-399.7

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 39 Logistic regression with interaction effect between local political exclusion and Temp dev

VARIABLES	(A137) Civil war onset	(A138) Civil war onset	(A139) Civil war onset	(A140) Civil war onset
Temp dev	-0.182	-0.171		
	(0.108)	(0.224)		
Exclusion x Temp dev		-0.016		
		(0.221)		
Temp dev (t-1)			-0.058	-0.029
			(0.156)	(0.185)
Exclusion x Temp dev (t-1)				-0.041
				(0.226)
Local political exclusion (t-1)	1.353*	1.356*	1.358*	1.372*
	(0.604)	(0.603)	(0.600)	(0.622)
Border distance (ln)	-1.421	-1.420	-1.446	-1.444*
	(0.737)	(0.738)	(0.742)	(0.737)
Capital distance (ln)	0.047	0.048	-0.026	-0.019
	(2.074)	(2.070)	(2.077)	(2.062)
Irrigated cropland (>10%)	-0.795*	-0.794*	-0.793*	-0.791*
	(0.372)	(0.372)	(0.368)	(0.368)
Rainfed cropland (>10%)	-0.708*	-0.708*	-0.706*	-0.704*
	(0.278)	(0.277)	(0.275)	(0.274)
Local population (ln)	0.463**	0.463**	0.464**	0.464**
	(0.093)	(0.093)	(0.094)	(0.093)
Democracy (t-1)	0.188*	0.188*	0.185*	0.185*
	(0.077)	(0.078)	(0.076)	(0.076)
GDP/capita (ln, t-1)	-0.457	-0.457	-0.457	-0.455
	(0.275)	(0.275)	(0.273)	(0.270)
Year	0.045*	0.045*	0.042	0.042
	(0.021)	(0.021)	(0.022)	(0.022)
Local conflict history	3.263**	3.262**	3.268**	3.269**
	(0.549)	(0.549)	(0.564)	(0.564)
Constant	-97.665*	-97.639*	-91.388*	-91.299*
	(39.828)	(39.906)	(42.550)	(42.379)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-399.0	-399.0	-400.4	-400.3

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 40 Logistic regression with interaction effect between local political exclusion and Drought

VARIABLES	(A141) Civil war onset	(A142) Civil war onset	(A143) Civil war onset	(A144) Civil war onset
Drought	-0.046	-0.665		
	(0.296)	(0.713)		
Exclusion x Drought		0.817		
		(0.846)		
Drought (t-1)			0.338	0.589
			(0.239)	(0.368)
Exclusion x Drought (t-1)				-0.386
				(0.460)
Local political exclusion (t-1)	1.356*	1.242*	1.367*	1.464*
	(0.600)	(0.599)	(0.595)	(0.631)
Border distance (ln)	-1.461*	-1.480*	-1.435	-1.436
	(0.742)	(0.740)	(0.748)	(0.748)
Capital distance (ln)	-0.041	-0.046	0.101	0.072
	(2.119)	(2.133)	(2.083)	(2.081)
Irrigated cropland (>10%)	-0.800*	-0.817*	-0.795*	-0.798*
	(0.381)	(0.380)	(0.379)	(0.375)
Rainfed cropland (>10%)	-0.709**	-0.710*	-0.719*	-0.722**
	(0.275)	(0.278)	(0.281)	(0.280)
Local population (ln)	0.465**	0.466**	0.464**	0.466**
	(0.094)	(0.095)	(0.094)	(0.094)
Democracy (t-1)	0.185*	0.183*	0.189*	0.189*
	(0.076)	(0.076)	(0.076)	(0.076)
GDP/capita (ln, t-1)	-0.458	-0.459	-0.458	-0.461
	(0.278)	(0.281)	(0.279)	(0.277)
Year	0.040	0.041	0.041*	0.041*
	(0.021)	(0.021)	(0.021)	(0.021)
Local conflict history	3.258**	3.254**	3.244**	3.240**
	(0.559)	(0.565)	(0.558)	(0.557)
Constant	-88.547*	-88.970*	-90.767*	-90.760*
	(39.395)	(39.798)	(39.955)	(39.794)
Observations	23,356	23,356	23,356	23,356
Log Likelihood	-400.5	-400.0	-399.8	-399.6

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 41 Logistic regression on 'most likely' subsample

VARIABLES	(A145) Civil war onset	(A146) Civil war onset	(A147) Civil war onset	(A148) Civil war onset	(A149) Civil war onset	(A150) Civil war onset
Precipitation		-0.171				
		(0.234)				
Temperature		0.099**				
		(0.034)				
Precipitation (t-1)			-0.048			
			(0.148)			
Temperature (t-1)			0.090**			
			(0.032)			
Prec growth (t-1 to t)				-0.253		
				(0.618)		
Prec growth (t-2 to t-1)				0.592		
				(0.501)		
Temp growth (t-1 to t)				-0.094		
				(1.028)		
Temp growth (t-2 to t-1)				-0.880		
				(0.867)		
Prec dev					-0.270	
					(0.164)	
Prec dev (t-1)					-0.028	
					(0.137)	
Temp dev					-0.237	
					(0.144)	
Temp dev (t-1)					-0.148	
					(0.353)	
Drought						0.148
						(0.256)
Drought (t-1)						-0.009
						(0.274)
Border distance (ln)	-2.114*	-2.035**	-2.001*	-2.104*	-2.036*	-2.127*
	(0.997)	(0.761)	(0.782)	(0.963)	(0.836)	(0.983)
Capital distance (ln)	-0.328	-1.376	-1.183	-0.455	-0.471	-0.350
	(3.265)	(2.972)	(3.018)	(3.372)	(3.246)	(3.293)
Irrigated cropland (>10%)	-1.110	-1.168	-1.135	-1.099	-1.168	-1.108
	(0.605)	(0.634)	(0.647)	(0.605)	(0.708)	(0.604)
Rainfed cropland (>10%)	-0.430	-0.379	-0.425	-0.451	-0.431	-0.426
	(0.357)	(0.389)	(0.370)	(0.391)	(0.381)	(0.360)
Local population (ln)	0.373**	0.222	0.234	0.383**	0.370**	0.375**
	(0.114)	(0.142)	(0.141)	(0.118)	(0.111)	(0.114)
Democracy (t-1)	0.237*	0.179	0.178	0.229*	0.239*	0.236*
	(0.109)	(0.108)	(0.109)	(0.107)	(0.109)	(0.108)
GDP/capita (ln, t-1)	-1.006**	-1.011**	-1.001**	-1.013**	-1.038**	-1.007**
	(0.357)	(0.274)	(0.275)	(0.347)	(0.355)	(0.354)
Year	0.086**	0.086**	0.086**	0.088**	0.097**	0.087**
	(0.026)	(0.019)	(0.019)	(0.025)	(0.031)	(0.025)
Local conflict history	3.257**	2.807**	2.790**	3.335**	3.368**	3.264**
	(0.626)	(0.411)	(0.404)	(0.641)	(0.618)	(0.633)
Constant	-172.757**	-171.425**	-170.720**	-175.793**	-194.325**	-174.141**
	(46.645)	(34.639)	(33.779)	(45.729)	(56.652)	(45.887)
Observations	7,674	7,674	7,674	7,674	7,674	7,674
Log Likelihood	-213.9	-205.4	-205.8	-212.4	-209.7	-213.9

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 42 Fixed-effects OLS

VARIABLES	(A151) Civil war onset	(A152) Civil war onset	(A153) Civil war onset	(A154) Civil war onset	(A155) Civil war onset	(A156) Civil war onset
Precipitation	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Precipitation (t-1)	0.003*	0.003*	0.003*	0.003*	0.003*	0.003*
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Temperature	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Temperature (t-1)	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Year		<0.001*	<0.001**	<0.001**	<0.001**	<0.001**
		(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Local population (ln)		<0.001**	<0.001*	<0.001*	0.001**	0.001**
		(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)
GDP/capita (ln, t-1)			-0.002**	-0.002**	-0.002*	-0.002*
			(0.001)	(0.001)	(0.001)	(0.001)
Democracy (t-1)				<0.001	<0.001	<0.001
				(<0.001)	(<0.001)	(<0.001)
Local political exclusion (t-1)					0.005**	0.005**
					(0.001)	(0.001)
Local conflict history						0.001
						(0.001)
Constant	0.003**	-0.080*	-0.393**	-0.388**	-0.330**	-0.325**
	(0.001)	(0.037)	(0.120)	(0.120)	(0.120)	(0.121)
Observations	30,845	30,833	30,812	30,812	30,734	30,734
R-squared	<0.001	0.001	0.001	0.001	0.002	0.002
Number of Countries	27	27	26	26	26	26

Robust standard errors in parentheses; ** p<0.01, * p<0.05

Table 43 Local civil war onset limited to 1000+ battle-related deaths

VARIABLES	(A157) Civil war onset	(A158) Civil war onset	(A159) Civil war onset	(A160) Civil war onset	(A161) Civil war onset	(A162) Civil war onset
Precipitation					-0.001	-0.001
					(0.001)	(0.001)
Temperature					0.001	0.001
					(0.001)	(0.001)
Precipitation (t-1)					>-0.001	>-0.001
					(<0.001)	(<0.001)
Temperature (t-1)					<0.001	<0.001
					(<0.001)	(<0.001)
Prec growth (t-1 to t)		-0.756				
		(0.810)				
Prec growth (t-2 to t-1)		0.639**				
		(0.238)				
Temp growth (t-1 to t)		-1.405*				
		(0.624)				
Temp growth (t-2 to t-1)		-1.036				
		(0.730)				
Prec dev			-0.191			
			(0.235)			
Prec dev (t-1)			0.156			
			(0.137)			
Temp dev			-0.248			
			(0.165)			
Temp dev (t-1)			0.080			
			(0.223)			
Drought				0.135		
				(0.378)		
Drought (t-1)				0.430		
				(0.394)		
Border distance (ln)	0.064	0.068	0.060	0.090		
	(1.035)	(0.997)	(0.995)	(1.032)		
Capital distance (ln)	-1.334	-1.519	-1.478	-1.145		
	(2.347)	(2.217)	(2.263)	(2.374)		
Irrigated cropland (>10%)	-0.771	-0.765	-0.757	-0.763		
	(0.513)	(0.499)	(0.507)	(0.517)		
Rainfed cropland (>10%)	-0.583	-0.569	-0.573	-0.590		
	(0.386)	(0.374)	(0.395)	(0.391)		
Local population (ln)	0.378*	0.383**	0.379*	0.379*		
	(0.148)	(0.145)	(0.149)	(0.150)		
Local political exclusion (t-1)	1.305*	1.312*	1.301*	1.312*		
	(0.582)	(0.590)	(0.590)	(0.577)		
Democracy (t-1)	0.124	0.119	0.123	0.129		
	(0.085)	(0.088)	(0.087)	(0.086)		
GDP/capita (ln, t-1)	-0.435	-0.476	-0.446	-0.435		
	(0.288)	(0.295)	(0.281)	(0.280)		
Year	0.033	0.037	0.037	0.034	<0.001	
	(0.021)	(0.022)	(0.025)	(0.020)	(<0.001)	
Local conflict history	2.795**	2.873**	2.809**	2.773**		
	(0.652)	(0.645)	(0.640)	(0.651)		
Constant	-72.722	-80.825	-81.944	-75.628*	-0.005	0.002**
	(39.284)	(42.218)	(46.737)	(38.370)	(0.025)	(0.001)
Observations	23,356	23,356	23,356	23,356	31,377	31,377
R-Squared					<0.001	0.004
Log Likelihood	-257.6	-253.6	-255.6	-256.8		
Number of Countries					27	27

Robust standard errors in parentheses; ** p<0.01, * p<0.05